

EXAM 7 – SPRING 2011

16. (2.5 points)

An insurance carrier wants to allocate capital by line of business using an appropriate risk measure to do so. The carrier writes two lines as follows:

- Line A is subject to catastrophe losses.
- Line B is excess casualty.

a. (1 point)

For each line, provide a moment-based or tail-based risk measure that would not be appropriate to use to allocate capital. Explain your answer.

b. (1 point)

For each line, provide a moment-based or tail-based risk measure that would be appropriate to use to allocate capital. Explain your answer.

c. (0.5 point)

Explain whether a distortion measure for Line A would make more sense than a moment-based or tail-based risk measure when allocating capital.

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Sample Solutions
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Solution 1

a)

Line A – VAR would not be appropriate as it is not coherent (fails subadditivity). Also VAR is a single percentile of the loss distribution and does not describe the risk in the tail

Line B – Standard deviation would not be appropriate as it does not work for skewed loss distributions. Also, standard deviation considers upside and downside where for xs casualty, we're more concerned with downside.

b)

Line A – TVAR would be more appropriate as it describes the expected risk in the whole tail. TVAR is also a coherent risk measure.

Line B – RTVAR (TVAR + conditional standard deviation) risk in the tail increases for higher probabilities which is representative of the inherent uncertainty for extreme casualty losses. RTVAR is also non-linear for exceedance which is more reasonable for xs casualty where ALAE is more prevalent and less likely to be proportional to losses.

c)

A distortion measure for allocating capital would make sense for Line A as it shifts significant probability towards the tail and so gives consistent charges across the board to high-level catastrophe covers as well as to standard coverages, all with the same parameterization.